INCH-POUND
MIL-DTL-16377/56A(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/56(SHIPS)
31 August 1971

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; 120 VOLTS, MACHINE TOOL, PORTABLE, FLOATING ADJUSTMENT, SYMBOL 266.2

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 2 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

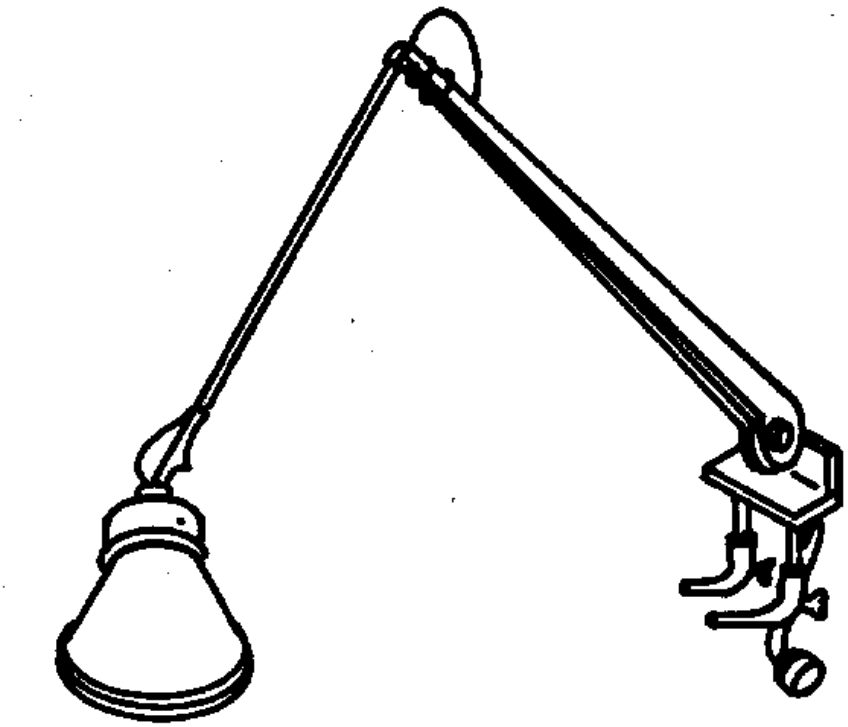


FIGURE 1. Typical view, machine tool light, symbol 266.2.

AMSC N/A FSC 6230

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:51Z
Check the source to verify that this is the current version before use.

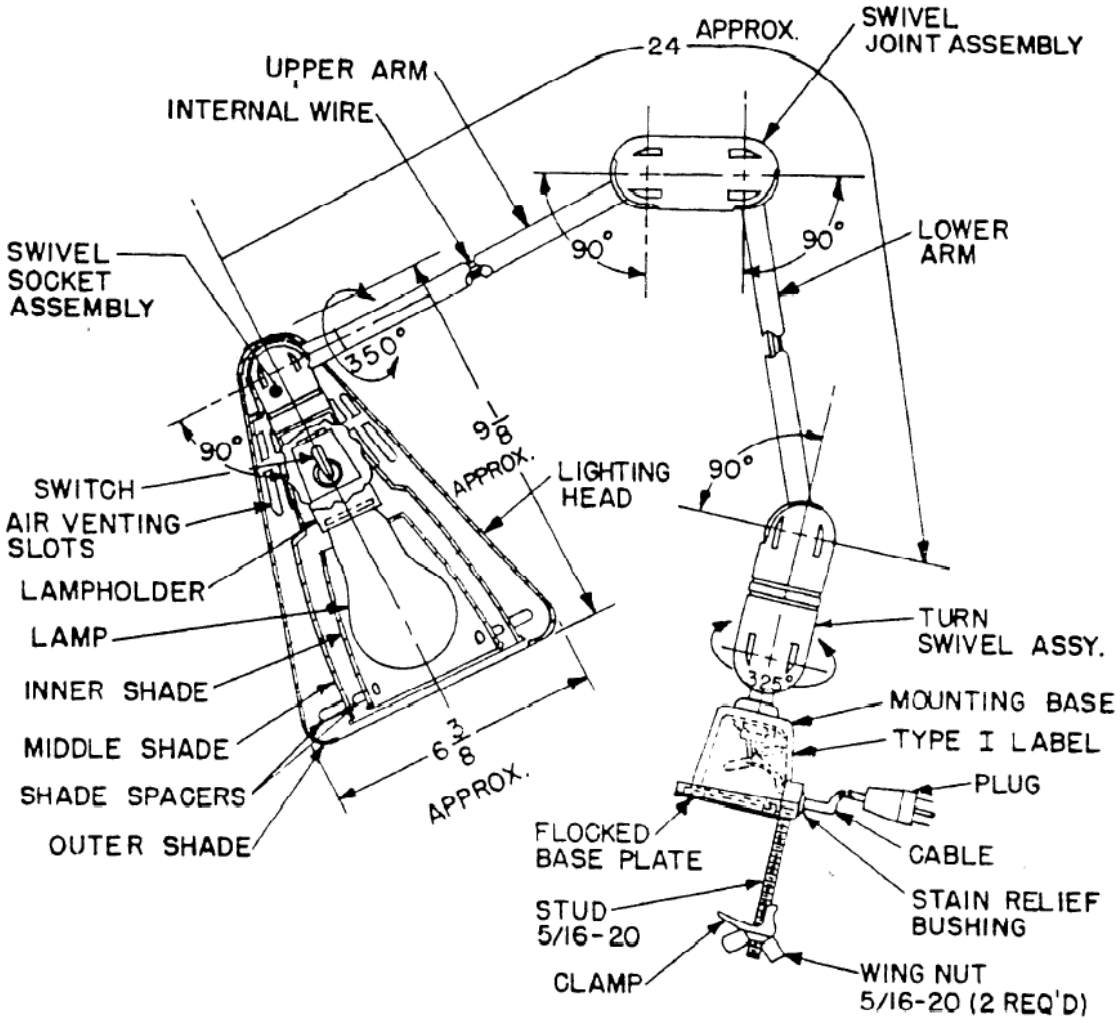


FIGURE 2. Section views, machine tool light, symbol 266.2.

REQUIREMENTS

CLASSIFICATION

Type: II or III
Class: 1

SYMBOL NUMBER

266.2: Light, 120 volts, machine tool, portable, floating adjustment.

DESIGN

Electrical rating: 100 watts for incandescent, 50 watts for solid-state lighting, 120 volts.
Dimensions and configuration: See figures 1 and 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:51Z
Check the source to verify that this is the current version before use.

Mounting base:	Material for mounting base shall be either zinc or aluminum. The base shall be not less than 3 inches wide of a universal type designed for vertical and horizontal mounting. The base shall include two removable auxiliary "C" clamps with an opening of not less than 2 inches and a throat depth of not less than 1¾ inches.
Arms:	Material for arms shall be steel tubing.
Lighting head:	The lighting head shall have three aluminum shades totally shielding a lamp, as shown on figure 2. Air venting slots shall be provided on the outer shade for cooling purposes. The shades shall be separated using nylon spacers.
Lamp:	The light shall be designed to accommodate a 100-watt, 120-volt, medium screw, inside frosted, rough service incandescent lamp in accordance with W-L-101/13 or an equivalent 120-volt solid-state lamp that provides equal or greater lumen output. (Lamp not furnished.)
Switch:	The switch shall be toggle, double pole, single throw, conforming to MS16569-1.
Cable and connector plug:	The light shall be equipped with 9 feet ±2 inches of cable, type SJC, AWG size 18, three-conductor (white, black, red) in accordance with UL 62. A three-conductor grounded plug shall be molded to end of cable in accordance with UL 817, type 1 assembly. All metal parts of the light shall be grounded to the red conductor and to the ground pin of the plug.
Internal wiring and grounding:	All internal wiring shall be routed within the tubular arms. Means shall be provided for positive ground connection within light for ground wire of supply cable. Positive grounding of all metal parts exposed to touch shall be maintained throughout entire light. Grounding between the outer or inner shades and ground terminal on plug shall meet the grounding requirements of MIL-DTL-16377.
Swivel joints:	The base, arms, and lighting head shall be connected with friction swivel joints that shall provide rotation and movement as shown on figure 2. The swivel joints shall provide sufficient friction to retain the lighting head within the tolerance specified in the vibration test.
Finish:	The finish shall be in accordance with MIL-DTL-16377, except that the inner shade shall be painted with a baked-on, high-temperature-resistant white enamel.
Enclosure effectiveness:	Totally enclosed in accordance with MIL-STD-108.
Weight:	5 pounds maximum.
Label:	One type I label shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figure 2.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:51Z
Check the source to verify that this is the current version before use.

TABLE I. First article and comparison inspection.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Grounding circuit	3.6.13.3	4.8.11	---
Photometrics	3.7.1	4.8.6	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---
Continuity of grounding	3.6.13.4	4.8.12	---

NOTE:

1/ The light shall be mounted as follows for this test:
(a) The lower arm shall make a 60-degree angle with the horizontal.
(b) The upper arm shall be parallel to the horizontal.
(c) The lighting head shall make a 45-degree angle with the upper arm facing downward.
Test shall be considered satisfactory if lighting head moves not more than 3 inches from placed position. Breakage of any part shall be cause for rejection.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6230-2023-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:51Z
Check the source to verify that this is the current version before use.